

数列

【题 1】设 $a_n = \left[\frac{(2n-1)!!}{(2n)!!} \right]^2$, 证明: $\{a_n\}$ 收敛于 0.

证明: 在不使用斯特林公式的情况下, 对任意 $n \geq 1$, 有

$$\prod_{k=1}^n \frac{(2k-1)(2k+1)}{(2k)^2} = \frac{1 \cdot 3^2 \cdot 5^2 \cdots (2n-1)^2(2n+1)}{2^2 \cdot 4^2 \cdots (2n)^2} = (2n+1)a_n,$$

又对每个 $k \geq 1$, 有 $0 < \frac{(2k-1)(2k+1)}{(2k)^2} = 1 - \frac{1}{4k^2} < 1$,

故 $0 < (2n+1)a_n < 1$, 从而 $0 < a_n < \frac{1}{2n+1} \rightarrow 0$,

由夹逼准则, $\lim_{n \rightarrow \infty} a_n = 0$.

【题 2】 $a_1 > 0$, $a_{n+1} = a_n + \frac{1}{a_n}$ 证明: $\lim_{n \rightarrow \infty} \frac{a_n}{\sqrt{2n}} = 1$

证明: 易得 $a_n > 0$, 且 $a_{n+1} > a_n$,

若 $\{a_n\}$ 收敛, 设收敛于 a , $a \in \mathbb{R}$, 则 $a = a + \frac{1}{a}$, 无解,

若 $\{a_n\}$ 有上界, 则由单调收敛定理必收敛, 与上面的结论矛盾.

故 $\{a_n\}$ 无上界; 又 $\{a_n\}$ 单调递增, 所以 $a_n \rightarrow +\infty$, 则 $\lim_{n \rightarrow \infty} \frac{1}{a_n} = 0$,

$$\therefore \lim_{n \rightarrow \infty} \frac{a_{n+1}^2 - a_n^2}{2(n+1) - 2n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{a_n^2} + 2}{2} = 1,$$

且 $b_n = 2n$ 为严格单调递增且发散到 $+\infty$ 的数列,

由 Stolz 定理, $\lim_{n \rightarrow \infty} \frac{a_n^2}{2n} = 1$,

$$\therefore \lim_{n \rightarrow \infty} \frac{a_n}{\sqrt{2n}} = 1.$$

一点思考: 关于为什么要平方, 我的思考是: 若有 $\lim_{n \rightarrow \infty} (b_{n+1} - b_n) = 0$,

则由 Stolz 公式, $\lim_{n \rightarrow \infty} \frac{b_n}{n} = 0$, 而这里 $a_{n+1} - a_n \rightarrow 0$,

但是题目要求 $\frac{a_n}{\sqrt{2n}} \rightarrow 1$, 这说明 $a_{n+1} - a_n \rightarrow 0$ 的条件不够强,

应该研究 $b_n = a_n^2$, 研究 $b_{n+1} - b_n$.

【题 3】 设 $0 < x_i \leq \frac{1}{2} \quad i = 1, 2, \dots, n$, 证明:

$$\frac{\prod_{i=1}^n x_i}{\left(\sum_{i=1}^n x_i\right)^n} \leq \frac{\prod_{i=1}^n (1 - x_i)}{\left[\sum_{i=1}^n (1 - x_i)\right]^n}$$

证明: $n = 1$ 时, 上式显然成立,

$$n = 2 \text{ 时, 即证 } \frac{x_1 x_2}{(x_1 + x_2)^2} \leq \frac{(1 - x_1)(1 - x_2)}{(2 - x_1 - x_2)^2},$$

$$\text{即证 } x_1 x_2 [4 + (x_1 + x_2)^2 - 4(x_1 + x_2)] \leq (x_1 + x_2)^2 [1 + x_1 x_2 - (x_1 + x_2)],$$

$$\text{即证 } 4x_1 x_2 [1 - (x_1 + x_2)] \leq (x_1 + x_2)^2 [1 - (x_1 + x_2)],$$

$$\because 1 - (x_1 + x_2) \geq 0, \text{ 即证 } 4x_1 x_2 \leq (x_1 + x_2)^2 \quad \text{易得上式成立,}$$

下面使用向前向后证明法,

向后: $n \rightarrow 2n$,

$$\left\{ \begin{array}{l} \frac{\prod_{i=1}^n x_i}{\frac{n}{(\sum_{i=1}^n x_i)^n}} \leq \frac{\prod_{i=1}^n (1-x_i)}{\frac{n}{[\sum_{i=1}^n (1-x_i)]^n}} \textcircled{1} \\ \frac{\prod_{i=n+1}^{2n} x_i}{\frac{2n}{(\sum_{i=n+1}^{2n} x_i)^n}} \leq \frac{\prod_{i=n+1}^{2n} (1-x_i)}{\frac{2n}{[\sum_{i=n+1}^{2n} (1-x_i)]^n}} \textcircled{2} \end{array} \right.$$

$$\textcircled{1} \times \textcircled{2} : \frac{\prod_{i=1}^{2n} x_i}{(\sum_{i=1}^n x_i \cdot \sum_{i=n+1}^{2n} x_i)^n} \leq \frac{\prod_{i=1}^{2n} (1-x_i)}{[\sum_{i=1}^n (1-x_i) \cdot \sum_{i=n+1}^{2n} (1-x_i)]^n}$$

$$\text{设 } A = \sum_{i=1}^n x_i \quad B = \sum_{i=n+1}^{2n} x_i \quad \text{则 } A \leq \frac{n}{2}, B \leq \frac{n}{2},$$

$$\therefore \frac{\prod_{i=1}^{2n} x_i}{\prod_{i=1}^{2n} (1-x_i)} \leq \frac{(AB)^n}{(n-A)^n(n-B)^n}, \quad \text{令 } A' = \frac{A}{n}, B' = \frac{B}{n},$$

$$\therefore \frac{(AB)^n}{(n-A)^n(n-B)^n} = \left[\frac{A'B'}{(1-A')(1-B')} \right]^n \quad \text{由 } n=2 \text{ 情形可知,}$$

$$\left[\frac{A'B'}{(1-A')(1-B')} \right]^n \leq \frac{(A'+B')^{2n}}{(2-A'-B')^{2n}} = \frac{(A+B)^{2n}}{(2n-A-B)^{2n}},$$

$$\text{即 } \frac{\prod_{i=1}^{2n} x_i}{\prod_{i=1}^{2n} (1-x_i)} \leq \frac{(\sum_{i=1}^{2n} x_i)^{2n}}{(2n - \sum_{i=1}^{2n} x_i)^{2n}} = \frac{(\sum_{i=1}^{2n} x_i)^{2n}}{[\sum_{i=1}^{2n} (1-x_i)]^{2n}},$$

$$\text{向前: } n \rightarrow n-1, \text{ 假设对 } n \text{ 成立, } \therefore \frac{1}{n-1} \sum_{i=1}^{n-1} x_i \leq \frac{1}{2} \quad \text{由 } x_n \text{ 的任意性,}$$

$$\text{不妨取 } x_n = \frac{1}{n-1} \sum_{i=1}^{n-1} x_i,$$

$$\text{设 } S_n = \sum_{i=1}^n x_i, \quad \text{则 } x_n = \frac{S_{n-1}}{n-1}, \quad S_n = S_{n-1} + x_n = S_{n-1} \cdot \frac{n}{n-1},$$

$$\therefore \frac{\prod_{i=1}^n x_i}{(S_n)^n} \leq \frac{\prod_{i=1}^n (1-x_i)}{(n-S_n)^n},$$

$$\therefore \frac{\prod_{i=1}^{n-1} x_i \cdot \frac{S_{n-1}}{n-1}}{S_{n-1}^n \cdot \frac{n^n}{(n-1)^n}} \leq \frac{\prod_{i=1}^{n-1} (1-x_i) \cdot (1 - \frac{S_{n-1}}{n-1})}{n^n (1 - \frac{S_{n-1}}{n-1})^n},$$

$$\therefore \frac{\prod_{i=1}^{n-1} x_i}{(\frac{S_{n-1}}{n-1})^{n-1}} \leq \frac{\prod_{i=1}^{n-1} (1-x_i)}{(1 - \frac{S_{n-1}}{n-1})^{n-1}},$$

$$\therefore \frac{\prod_{i=1}^{n-1} x_i}{(\sum_{i=1}^{n-1} x_i)^{n-1}} \leq \frac{\prod_{i=1}^{n-1} (1-x_i)}{\left[\sum_{i=1}^{n-1} (1-x_i)\right]^{n-1}},$$

又 $n=2$ 时命题成立,

$$\text{由向前-向后归纳法可知 } \frac{\prod_{i=1}^n x_i}{(\sum_{i=1}^n x_i)^n} \leq \frac{\prod_{i=1}^n (1-x_i)}{\left[\sum_{i=1}^n (1-x_i)\right]^n}.$$

【题 4】 设 $a_n = \sqrt{1 + \sqrt{2 + \sqrt{\dots + \sqrt{n}}}}$ 证明: $\{a_n\}$ 收敛

证明: 易得 $\{a_n\}$ 单调递增,

对固定的 n , 定义 $x_{n,n} = \sqrt{n}$, $x_{k,n} = \sqrt{k + x_{k+1,n}}$ ($k = n-1, \dots, 1$), 则 $a_n = x_{1,n}$,

下面用反向归纳法证明 $x_{k,n} \leq \sqrt{k} + 1$ ($1 \leq k \leq n$),

当 $k=n$ 时, $x_{n,n} = \sqrt{n} \leq \sqrt{n} + 1$. 若 $x_{k+1,n} \leq \sqrt{k+1} + 1$,

则由 $\sqrt{k+1} \leq 2\sqrt{k}$ 可得 $x_{k,n} \leq \sqrt{k + \sqrt{k+1} + 1} \leq \sqrt{k + 2\sqrt{k} + 1} = \sqrt{k} + 1$,

故由反向归纳法, $a_n = x_{1,n} \leq 2$, 即 $\{a_n\}$ 有上界.

又 $\{a_n\}$ 单调递增, 所以由单调收敛定理, $\{a_n\}$ 收敛.

【题 5】 设数列 $\{a_n\}_{n=0}^{\infty}$ 满足 $\lim_{n \rightarrow \infty} a_n = a$, 证明 $\lim_{n \rightarrow \infty} \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k} a_k = a$

$$\text{证明: } \because 2^n = \sum_{k=0}^n \binom{n}{k},$$

$$\therefore \left| \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k} a_k - a \right| = \left| \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k} (a_k - a) \right| \leq \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k} |a_k - a|,$$

$$\because \lim_{n \rightarrow \infty} a_n = a \quad \therefore \text{对 } \forall \varepsilon > 0, \exists N_1 \text{ s.t. } n > N_1 \text{ 时, } |a_n - a| < \varepsilon,$$

$$\therefore \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k} |a_k - a| = \frac{1}{2^n} \sum_{k=0}^{N_1} \binom{n}{k} |a_k - a| + \frac{1}{2^n} \sum_{k=N_1+1}^n \binom{n}{k} |a_k - a|$$

$$< \frac{1}{2^n} \sum_{k=0}^{N_1} \binom{n}{k} |a_k - a| + \frac{\varepsilon}{2^n} \sum_{k=N_1+1}^n \binom{n}{k}$$

$$< \frac{1}{2^n} \sum_{k=0}^{N_1} \binom{n}{k} |a_k - a| + \varepsilon,$$

$$\text{又 } \binom{n}{k} = \frac{n!}{k!(n-k)!} \leq \frac{n!}{(n-k)!} \leq n^k, \text{ 令 } M = \max\{|a_0 - a|, |a_1 - a|, \dots, |a_{N_1} - a|\},$$

$$\therefore \frac{1}{2^n} \sum_{k=0}^{N_1} \binom{n}{k} |a_k - a| \leq \frac{M}{2^n} \sum_{k=0}^{N_1} n^k < \frac{M \cdot n^{N_1+1}}{2^n},$$

$$\because \lim_{n \rightarrow \infty} \frac{n^\alpha}{2^n} = 0 \quad \therefore \forall \varepsilon, \exists N_2, \text{ s.t. } n > N_2 \text{ 时, } \frac{n^{N_1+1}}{2^n} < \varepsilon,$$

$$\therefore \forall \varepsilon > 0, \exists N = \max\{N_1, N_2\}, n > N \text{ 时, } \left| \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k} a_k - a \right| < (M+1)\varepsilon,$$

$$\therefore \lim_{n \rightarrow \infty} \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k} a_k = a.$$

【题 6】 设 a_n 满足条件 $0 < a_1 < 1, a_{n+1} = a_n(1 - a_n)$, 证明: $\lim_{n \rightarrow \infty} n a_n = 1$

$$\text{证明: } \because a_{n+1} - a_n = -a_n^2 \leq 0, \quad \therefore \{a_n\} \text{ 单调递减,}$$

由数学归纳法可得 $0 < a_n < 1$, $\therefore \{a_n\}$ 收敛,

设 $\lim_{n \rightarrow \infty} a_n = \alpha$, 其中 $0 \leq \alpha \leq 1$, $\therefore \alpha = \alpha(1 - \alpha) \Rightarrow \alpha = 0$,

$$\text{又 } \frac{1}{a_{n+1}} = \frac{1}{a_n(1 - a_n)} = \frac{1}{a_n} + \frac{1}{1 - a_n}, \quad \therefore \frac{1}{a_{n+1}} - \frac{1}{a_n} = \frac{1}{1 - a_n},$$

$$\therefore \lim_{n \rightarrow \infty} \frac{(n+1) - n}{\frac{1}{a_{n+1}} - \frac{1}{a_n}} = \lim_{n \rightarrow \infty} (1 - a_n) = 1,$$

又 $\frac{1}{a_n}$ 为严格单调递增的正无穷大量,

$$\therefore \lim_{n \rightarrow \infty} \frac{n}{\frac{1}{a_n}} = 1, \text{ 即 } \lim_{n \rightarrow \infty} na_n = 1.$$

注: 本题与题 2 类似, 直接用 Stolz 公式并不好做, 取倒数的动机为如果 $\lim_{n \rightarrow \infty} na_n = 1$

这就说明 a_n 的衰减速度与 $\frac{1}{n}$ 相同, 则 $b_n = \frac{1}{a_n}$ 就是线性增长, 便于研究.

【题 7】 证明: 自然对数的底 e 是无理数

证明: 假设 $e \in \mathbb{Q}$, 则 $\exists p, q \in \mathbb{N}^+$, s.t. $e = \frac{p}{q}$,

$$\text{设 } \varepsilon_n = e - (1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!}),$$

先证明一个引理: $0 < \varepsilon_n < \frac{1}{n!n}$, 其中 $n \geq 1$. 由级数展开及几何级数求和,

$$n!\varepsilon_n = \sum_{j=1}^{\infty} \frac{1}{(n+1)(n+2)\cdots(n+j)} < \sum_{j=1}^{\infty} \frac{1}{(n+1)^j} = \frac{1}{n},$$

$$\therefore q \in \mathbb{N}^+, e = (1 + \frac{1}{1!} + \dots + \frac{1}{q!}) + \varepsilon_q = \frac{p}{q},$$

$$\text{两边同乘 } q!, \text{ 有 } p(q-1)! - q! \sum_{k=0}^q \frac{1}{k!} = q!\varepsilon_q,$$

等式左边为整数, 故 $q!\varepsilon_q \in \mathbb{Z}$. 另一方面, $0 < q!\varepsilon_q < \frac{1}{q} \leq 1$, 即 $0 < q!\varepsilon_q < 1$, 矛盾!

这就说明 e 是无理数, 得证.

【题 8】 设已知存在极限 $\lim_{n \rightarrow \infty} \frac{a_1 + a_2 + \dots + a_n}{n}$, 证明: $\lim_{n \rightarrow \infty} \frac{a_n}{n} = 0$

证明: 设 $b_n = \frac{a_1 + a_2 + \dots + a_n}{n}$, 由题意, 不妨设 $\{b_n\}$ 的极限为 l ,

$$\because a_n = nb_n - (n-1)b_{n-1} \quad (n \geq 2),$$

$$\therefore \lim_{n \rightarrow \infty} \frac{a_n}{n} = \lim_{n \rightarrow \infty} \frac{nb_n - (n-1)b_{n-1}}{n} = l - l = 0.$$

【题 9】设 $x_n = \frac{1}{n^2} \sum_{k=0}^n \ln \binom{n}{k}$, 求 $\lim_{n \rightarrow \infty} x_n$

证明: 由于 n^2 是单调递增的正无穷大量,

$$\text{由 Stolz 公式, } \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \frac{\sum_{k=0}^{n+1} \ln \binom{n+1}{k} - \sum_{k=0}^n \ln \binom{n}{k}}{2n+1}$$

$$= \lim_{n \rightarrow \infty} \frac{\sum_{k=0}^n \ln \frac{n+1}{n+1-k}}{2n+1}, \text{ 又由于 } 2n+1 \text{ 也是单调递增的正无穷大量,}$$

$$\text{再由 Stolz 公式, } \lim_{n \rightarrow \infty} \frac{\sum_{k=0}^n \ln \frac{n+1}{n+1-k}}{2n+1} = \lim_{n \rightarrow \infty} \frac{\sum_{k=0}^n \ln \frac{n+1}{n+1-k} - \sum_{k=0}^{n-1} \ln \frac{n}{n-k}}{2}$$

$$= \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n \ln \frac{n+1}{n+1-k} - \sum_{k=0}^{n-1} \ln \frac{n}{n-k}}{2}$$

$$= \lim_{n \rightarrow \infty} \frac{\sum_{k=0}^{n-1} \ln \frac{n+1}{n-k} - \sum_{k=0}^{n-1} \ln \frac{n}{n-k}}{2}$$

$$= \lim_{n \rightarrow \infty} \frac{n \ln \frac{n+1}{n}}{2},$$

$$\because \frac{1}{n+1} \leq \ln\left(1 + \frac{1}{n}\right) \leq \frac{1}{n}, \quad \therefore \frac{n}{n+1} \leq n \ln\left(1 + \frac{1}{n}\right) \leq 1,$$

$$\text{由夹逼准则, } \lim_{n \rightarrow \infty} n \ln\left(1 + \frac{1}{n}\right) = 1,$$

$$\therefore \lim_{n \rightarrow \infty} x_n = \frac{1}{2}.$$

【题 10】 设 $a_n > 0$, $\{\frac{a_{n+1} + a_{n+2}}{a_n}\}$ 收敛于 α , $\alpha > 2$, 证明: 数列 $\{a_n\}$ 没有上界

证明: 用反证法, 假设 $\{a_n\}$ 有上界, 又 $a_n > 0 \Rightarrow \{a_n\}$ 为有界序列,

所以 $\{a_n\}$ 上下极限存在, 记 $a = \overline{\lim}_{n \rightarrow \infty} a_n$, $b = \underline{\lim}_{n \rightarrow \infty} a_n$, 那么 $a \geq b \geq 0$,

① $a > 0$: $\exists \beta > 2, N_1$, s.t. $n > N_1$ 时, $a_{n+1} + a_{n+2} > \beta a_n$,

由上极限的保不等式性质, $2a \geq \beta a \Rightarrow \beta \leq 2$, 矛盾!

② $a = 0$: 同上有 $a_{n+1} + a_{n+2} > \beta a_n (\beta > 2, n > N_1)$,

记 $M_n = \sup_{k \geq n} a_k$, 则 $a_{n+1} \leq M_n, a_{n+2} \leq M_n \Rightarrow M_n > \frac{\beta}{2} a_n > a_n$,

又 $M_n = \max\{a_n, M_{n+1}\}$, $\therefore M_n = M_{n+1}, \forall n > N_1$,

又 $a = 0 \Rightarrow \lim_{n \rightarrow \infty} M_n = 0 \therefore \forall n > N_1$, 都有 $M_n = 0$, 这与 $a_n > 0$ 矛盾!

综上, $\{a_n\}$ 没有上界.

注意: 上下极限没有乘除法的四则运算性质, 所以即使是在 $a > b > 0$ 的情况下,
 $\alpha \neq \frac{2a}{b}, \alpha \neq \frac{2b}{a}$